

ToughTorq Bolt Tensioners - Spring Return Series

Automatic spring return | Broad thread coverage | Simultaneous multiple-tool tensioning

Model	Thread Range	Max Tension	Eff. Area	Stroke	Return
TT-ST1	M24â M27	63,396 lbf	2.92 in2	0.31 in	Spring
TT-ST2	M24â M36	101,614 lbf	4.67 in2	0.39 in	Spring
TT-ST3	M39â M48	150,847 lbf	6.94 in2	0.39 in	Spring
TT-ST4	M39â M48	226,832 lbf	10.43 in2	0.39 in	Spring
TT-ST5	M52â M68	345,756 lbf	15.89 in2	0.39 in	Spring
TT-ST6	M72â M90	571,914 lbf	26.3 in2	0.47 in	Spring
TT-ST7	M90â M100	722,986 lbf	33.23 in2	0.47 in	Spring

ToughTorq Bolt Tensioners - Load Return Series

Modular adaptor kits | Twin hydraulic ports | 21,756 psi / 150 MPa reference pressure where published

Model	Thread Range	Max Load	Pressure	Eff. Area	Stroke	Weight
TTGT1-LCB	M16-M30	50,425 lbf	21,756 psi	2.32 in2	0.39 in	6.6 lb
TTGT2-LCB	M30-M39	90,261 lbf	21,756 psi	4.15 in2	0.39 in	9 lb
TTGT3-LCB	M39-M52	172,856 lbf	21,756 psi	7.95 in2	0.39 in	15.4 lb
TTGT4-LCB	M52-M68	329,772 lbf	21,756 psi	15.16 in2	0.39 in	26.9 lb
TTGT5-LCB	M68-M80	508,383 lbf	21,756 psi	23.37 in2	0.39 in	41.2 lb
TTGT6-LCB	M80-M95	639,604 lbf	21,756 psi	29.41 in2	0.39 in	61.3 lb

ToughTorq Bolt Tensioners - Multistage Series

Compact high-capacity tensioning for restricted radial clearance

Model	Bolt	Max Pull	Pressure	Stroke	Eff.Area	OD	Weight
TTE-24	M24	69,016 lbf	21,756 psi	0.28 in	3.52 in2	2.32 in	11 lb
TTE-30	M30	103,637 lbf	21,756 psi	0.28 in	5.3 in2	2.83 in	14.6 lb
TTE-33	M33	133,761 lbf	21,756 psi	0.28 in	6.83 in2	3.11 in	17.6 lb
TTE-36	M36	152,870 lbf	21,756 psi	0.39 in	7.81 in2	3.33 in	23.6 lb
TTE-39	M39	179,847 lbf	21,756 psi	0.39 in	9.19 in2	3.6 in	28.2 lb
TTE-42	M42	207,274 lbf	21,756 psi	0.39 in	10.59 in2	3.88 in	35.3 lb
TTE-45	M45	244,143 lbf	21,756 psi	0.39 in	12.47 in2	4.21 in	46.3 lb
TTE-48	M48	291,352 lbf	21,756 psi	0.39 in	14.89 in2	4.57 in	48.5 lb
TTE-56	M56	373,183 lbf	21,756 psi	0.39 in	19.06 in2	5.16 in	52.9 lb
TTE-64	M64	528,301 lbf	21,756 psi	0.39 in	29.14 in2	5.82 in	103.6 lb